

University of Central Punjab
Faculty of Information Technology
Object Oriented Programming
Fall 2025

Lab 13

Topic

- Classes in C++, access specifiers, objects, methods, STL Components

Objective

Making students familiarize with the concepts of STL Components.

CLO

Apply object-oriented programming principles to implement real-world problems.

Instructions:

- Indent your code.
- Comment your code.
- Use meaningful variable names.
- Plan your code carefully on a piece of paper before you implement it.
- Name of the program should be same as the task name. i.e. the first program should be Task_1.cpp

- **void main() is not allowed. Use int main()**
- **You are not allowed to use any built-in functions**
- **You are required to follow the naming conventions as follow:**
 - **Variables:** firstName; (no underscores allowed)
 - **Function:** getName(); (no underscores allowed)
 - **ClassName:** BankAccount (no underscores allowed)

Students are required to complete the following tasks in lab timings.

Question 1:

Create a class IntVector that has a private `std::vector<int>` member. Add a public method `addElement()` to insert integers 10, 20, 30, 40, and 50 into the vector, and another method `printElements()` that uses an iterator to print all elements.

Question 2:

Extend the IntVector class with a method `doubleElements()` that doubles each element in the vector using iterators.

Question 3:

Add a method `printElementsConst()` in the IntVector class that uses a **constant iterator** (`cbegin / cend`) to print elements without modifying them.

Question 4:

Add a method `removeEvens()` to the IntVector class that removes all even numbers from the vector using iterators.